I will be showing the before and after of how the TLS takes effect and changes what someone that's unauthorized sees with encryption vs without encryption. (MQTT)

- This below is a dashboard that is subscribed to the broker to get readings from the sensors

```
Anaconda Prompt - python "
  Pressure (downstream):    75.4 PSI
  Pressure differential:     7.9 PSI
  Flow rate:                42.8 gal/min

  Location:   main-building
  Device ID:  GM-HYDROLOGIC-01
  Time:       2026-04-21T05:47:58.284000+00:00
  Count:      #51
  Pressure (upstream):      81.9 PSI
  Pressure (downstream):    75.7 PSI
  Pressure differential:     6.2 PSI
  Flow rate:                39.6 gal/min

  Location:   kitchen
  Device ID:  GM-HYDROLOGIC-03
  Time:       2026-04-21T05:47:58.358186+00:00
  Count:      #51
  Pressure (upstream):      80.6 PSI
  Pressure (downstream):    77.9 PSI
  Pressure differential:     2.7 PSI
  Flow rate:                37.5 gal/min

  Location:   pool-wing
  Device ID:  GM-HYDROLOGIC-02
  Time:       2026-04-21T05:47:58.358298+00:00
  Count:      #51
```

- The reading below is me accessing the sensor readings to show how easy it is without TLS and Certificate Authorities to gain access to this crucial information
  - Command used `mosquitto_sub -t "#" -h`

```
Anaconda Prompt - mosquitt
38.428545+00:00", "counter": 71, "pressure_upstream": 82.4, "pressure_downstream": 77.5, "flow_rate": 41.0}
hydroficient/grandmarina/sensors/pool-wing/readings {"device_id": "GM-HYDROLOGIC-02", "location": "pool-wing", "timestamp": "2026-04-21T05
:48:38.428633+00:00", "counter": 71, "pressure_upstream": 83.9, "pressure_downstream": 77.7, "flow_rate": 39.3}
hydroficient/grandmarina/sensors/main-building/readings {"device_id": "GM-HYDROLOGIC-01", "location": "main-building", "timestamp": "2026-
04-21T05:48:40.342681+00:00", "counter": 72, "pressure_upstream": 81.1, "pressure_downstream": 74.1, "flow_rate": 41.7}
hydroficient/grandmarina/sensors/kitchen/readings {"device_id": "GM-HYDROLOGIC-03", "location": "kitchen", "timestamp": "2026-04-21T05:48:
40.429595+00:00", "counter": 72, "pressure_upstream": 81.5, "pressure_downstream": 77.1, "flow_rate": 41.4}
hydroficient/grandmarina/sensors/pool-wing/readings {"device_id": "GM-HYDROLOGIC-02", "location": "pool-wing", "timestamp": "2026-04-21T05
:48:40.430473+00:00", "counter": 72, "pressure_upstream": 81.9, "pressure_downstream": 76.9, "flow_rate": 38.0}
hydroficient/grandmarina/sensors/main-building/readings {"device_id": "GM-HYDROLOGIC-01", "location": "main-building", "timestamp": "2026-
04-21T05:48:42.345368+00:00", "counter": 73, "pressure_upstream": 82.4, "pressure_downstream": 75.3, "flow_rate": 37.7}
hydroficient/grandmarina/sensors/kitchen/readings {"device_id": "GM-HYDROLOGIC-03", "location": "kitchen", "timestamp": "2026-04-21T05:48:
42.431502+00:00", "counter": 73, "pressure_upstream": 81.5, "pressure_downstream": 74.1, "flow_rate": 38.5}
hydroficient/grandmarina/sensors/pool-wing/readings {"device_id": "GM-HYDROLOGIC-02", "location": "pool-wing", "timestamp": "2026-04-21T05
:48:42.431591+00:00", "counter": 73, "pressure_upstream": 83.6, "pressure_downstream": 76.3, "flow_rate": 42.2}
hydroficient/grandmarina/sensors/main-building/readings {"device_id": "GM-HYDROLOGIC-01", "location": "main-building", "timestamp": "2026-
04-21T05:48:44.346602+00:00", "counter": 74, "pressure_upstream": 83.5, "pressure_downstream": 77.1, "flow_rate": 41.9}
hydroficient/grandmarina/sensors/kitchen/readings {"device_id": "GM-HYDROLOGIC-03", "location": "kitchen", "timestamp": "2026-04-21T05:48:
44.432808+00:00", "counter": 74, "pressure_upstream": 81.6, "pressure_downstream": 76.2, "flow_rate": 40.5}
hydroficient/grandmarina/sensors/pool-wing/readings {"device_id": "GM-HYDROLOGIC-02", "location": "pool-wing", "timestamp": "2026-04-21T05
:48:44.434083+00:00", "counter": 74, "pressure_upstream": 81.2, "pressure_downstream": 77.5, "flow_rate": 41.8}
hydroficient/grandmarina/sensors/main-building/readings {"device_id": "GM-HYDROLOGIC-01", "location": "main-building", "timestamp": "2026-
04-21T05:48:46.348971+00:00", "counter": 75, "pressure_upstream": 82.4, "pressure_downstream": 74.9, "flow_rate": 37.9}
hydroficient/grandmarina/sensors/pool-wing/readings {"device_id": "GM-HYDROLOGIC-02", "location": "pool-wing", "timestamp": "2026-04-21T05
:48:46.435268+00:00", "counter": 75, "pressure_upstream": 83.8, "pressure_downstream": 75.5, "flow_rate": 41.2}
hydroficient/grandmarina/sensors/kitchen/readings {"device_id": "GM-HYDROLOGIC-03", "location": "kitchen", "timestamp": "2026-04-21T05:48:
46.435157+00:00", "counter": 75, "pressure_upstream": 83.4, "pressure_downstream": 77.7, "flow_rate": 38.3}
hydroficient/grandmarina/sensors/main-building/readings {"device_id": "GM-HYDROLOGIC-01", "location": "main-building", "timestamp": "2026-
04-21T05:48:48.352214+00:00", "counter": 76, "pressure_upstream": 82.9, "pressure_downstream": 75.4, "flow_rate": 42.7}
```

- Since this is only documentation and I won't be displaying large amounts of code here as it would just make the pages unreadable, I have linked the coding part in my GitHub, Here I will display configuration files/experiments
- After generating three different certificates (CA Root, Broker Server, Private Key) I opened up Mosquitto's TLS port from its unencrypted port 1883 -> 8883
- Cafile is for the root certificate, Certfile is for the broker clients certificate, Keyfile is the private key used to verify the clients broker (Verified by the Root CA)

mosquitto_tls.conf

```
# mosquitto_tls.conf
listener 8883 localhost

# TLS Certificate Configuration
cafile certs/ca.pem
certfile certs/server.pem
keyfile certs/server-key.pem

# Client certificate requirement
require_certificate false

# Authentication
allow_anonymous true

# Logging
log_type all
log_dest stdout
```

- I attempted a connection without TLS and was given an error mention that no connection could be established, I also verified that the port was open to 8883 and not 1883

```
(base) C:\Users\nsain\Documents\hydroficient-project>mosquitto_sub -h localhost -p 8883 -t "test/#"
Error: protocol error

(base) C:\Users\nsain\Documents\hydroficient-project>netstat -an | findstr "8883"
  TCP    127.0.0.1:8883         0.0.0.0:0              LISTENING
  TCP    [::1]:8883             [::]:0                 LISTENING

(base) C:\Users\nsain\Documents\hydroficient-project>
```

- After importing ssl and importing created certificates through a dictionary created by the name of TLS_CONFIG we have given our dashboard(subscriber) and the Mosquitto broker(publisher) to get authorized by the CA before being able to connect to the client.

```
1776801381: New client connected from ::1:13609 as auto-15864108-1D71-5130-0696-E6502FB3D12E (p4, c1, k60).
1776801381: No will message specified.
1776801381: Client auto-15864108-1D71-5130-0696-E6502FB3D12E negotiated TLSv1.3 cipher TLS_AES_256_GCM_SHA384
1776801381: Sending CONNACK to auto-15864108-1D71-5130-0696-E6502FB3D12E (0, 0)
1776801381: New client connected from ::1:13610 as auto-3E0CE087-E590-D92C-47BE-1B02D56297DE (p4, c1, k60).
1776801381: No will message specified.
1776801381: Client auto-3E0CE087-E590-D92C-47BE-1B02D56297DE negotiated TLSv1.3 cipher TLS_AES_256_GCM_SHA384
1776801381: Sending CONNACK to auto-3E0CE087-E590-D92C-47BE-1B02D56297DE (0, 0)
1776801381: New client connected from ::1:13611 as auto-5EE6AEE4-6053-AD65-1ECF-5ABBBE570C76 (p4, c1, k60).
1776801381: No will message specified.
1776801381: Client auto-5EE6AEE4-6053-AD65-1ECF-5ABBBE570C76 negotiated TLSv1.3 cipher TLS_AES_256_GCM_SHA384
1776801381: Sending CONNACK to auto-5EE6AEE4-6053-AD65-1ECF-5ABBBE570C76 (0, 0)
1776801381: Received PUBLISH from auto-15864108-1D71-5130-0696-E6502FB3D12E (d0, q0, r0, m0, 'hydroficient/grandmarina/s
ensors/main-building/readings', ... (200 bytes))
```

- Below is a full working TLS encrypted connection, the subscriber is receiving logs from the publisher and it's also displaying CONACK (new client connected) on Mosquitto

```
Anaconda Prompt - python s
  Location:  pool-wing
  Device ID: GM-HYDROLOGIC-02
  Time:      2026-04-21T20:33:03.365645+00:00
  Count:     #80
  ──────────────────────────────────────────
  Pressure (upstream):     82.0 PSI
  Pressure (downstream):   77.3 PSI
  Pressure differential:    4.7 PSI
  Flow rate:               42.8 gal/min

  ──────────────────────────────────────────
  Location:  main-building
  Device ID: GM-HYDROLOGIC-01
  Time:      2026-04-21T20:33:05.363442+00:00
  Count:     #81
  ──────────────────────────────────────────
  Pressure (upstream):     83.3 PSI
  Pressure (downstream):   77.2 PSI
  Pressure differential:    6.1 PSI
  Flow rate:               41.1 gal/min

  ──────────────────────────────────────────
  Location:  kitchen
  Device ID: GM-HYDROLOGIC-03
  Time:      2026-04-21T20:33:05.367066+00:00
  Count:     #81
  ──────────────────────────────────────────
  Pressure (upstream):     82.4 PSI
  Pressure (downstream):   76.7 PSI
  Pressure differential:    5.7 PSI
  Flow rate:               42.3 gal/min

  ──────────────────────────────────────────
  Location:  pool-wing
  Device ID: GM-HYDROLOGIC-02
  Time:      2026-04-21T20:33:05.367138+00:00
  Count:     #81
  ──────────────────────────────────────────
  Pressure (upstream):     81.4 PSI
  Pressure (downstream):   75.7 PSI
  Pressure differential:    5.7 PSI
  Flow rate:               41.4 gal/min
```

```
C:\Program Files\mosquitto\mosquitto.exe
1776803579: Received PUBLISH from auto-7E09804B-D9E0-D44E-431A-C613DF37AF4A (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/kitchen/readings', ... (195 bytes))
1776803579: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/k
itchen/readings', ... (195 bytes))
1776803581: Received PUBLISH from auto-224658AC-E010-ABA5-2871-0DB7837F7F2E (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/main-building/readings', ... (201 bytes))
1776803581: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/m
ain-building/readings', ... (201 bytes))
1776803581: Received PUBLISH from auto-ECF6533F-CA91-2769-33E1-A2CCE9F9290F (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/pool-wing/readings', ... (197 bytes))
1776803581: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/p
ool-wing/readings', ... (197 bytes))
1776803581: Received PUBLISH from auto-7E09804B-D9E0-D44E-431A-C613DF37AF4A (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/kitchen/readings', ... (195 bytes))
1776803581: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/k
itchen/readings', ... (195 bytes))
1776803583: Received PUBLISH from auto-224658AC-E010-ABA5-2871-0DB7837F7F2E (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/main-building/readings', ... (201 bytes))
1776803583: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/m
ain-building/readings', ... (201 bytes))
1776803583: Received PUBLISH from auto-7E09804B-D9E0-D44E-431A-C613DF37AF4A (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/kitchen/readings', ... (195 bytes))
1776803583: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/k
itchen/readings', ... (195 bytes))
1776803583: Received PUBLISH from auto-ECF6533F-CA91-2769-33E1-A2CCE9F9290F (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/pool-wing/readings', ... (197 bytes))
1776803583: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/p
ool-wing/readings', ... (197 bytes))
1776803585: Received PUBLISH from auto-224658AC-E010-ABA5-2871-0DB7837F7F2E (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/main-building/readings', ... (201 bytes))
1776803585: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/m
ain-building/readings', ... (201 bytes))
1776803585: Received PUBLISH from auto-7E09804B-D9E0-D44E-431A-C613DF37AF4A (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/kitchen/readings', ... (195 bytes))
1776803585: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/k
itchen/readings', ... (195 bytes))
1776803585: Received PUBLISH from auto-ECF6533F-CA91-2769-33E1-A2CCE9F9290F (d0, q0, r0, m0, 'hydroficient
/grandmarina/sensors/pool-wing/readings', ... (197 bytes))
1776803585: Sending PUBLISH to grand-marina-dashboard (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/p
ool-wing/readings', ... (197 bytes))
```

Phase 2: Penetration Testing

The connection is secure from what we see, but is it really secure? In the next 4 steps I will be Eavesdrops, Creating fake certificates, Checking security speed if it will slow down leak detection, and Stress testing how much the sensors can handle.

- **Experiment 1/4** I ran an eavesdropping command over port 1883/8883 to see if I was able to connect and receive the logs with TLS off/on, with TLS off the logs sent over were insecure VS. TLS on the eavesdropping terminal was unable to connect resulting in an Error: Protocol Error, Messages afterwards were sent securely.

```
(base) C:\Users\nsain\Documents\hydroficient-project\Project-04>python experiment_runner.py --mode publish --tls off --c
ount 5

==================================================
  Publishing 5 messages
  TLS: OFF
  Port: 1883
==================================================

Published message 1/5: pressure=82.5 PSI
Published message 2/5: pressure=83.5 PSI
Published message 3/5: pressure=84.5 PSI
Published message 4/5: pressure=85.5 PSI
Published message 5/5: pressure=86.5 PSI

Done publishing!
```

```
(base) C:\Users\nsain\Documents\hydroficient-project\Project-04>mosquitto_sub -h localhost -p 1883 -t "grandmarina/#" -v

grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 1, "pressure_psi": 82.5, "flow_rate_g
pm": 45.0, "timestamp": "11:09:49"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 2, "pressure_psi": 83.5, "flow_rate_g
pm": 45.0, "timestamp": "11:09:50"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 3, "pressure_psi": 84.5, "flow_rate_g
pm": 45.0, "timestamp": "11:09:51"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 4, "pressure_psi": 85.5, "flow_rate_g
pm": 45.0, "timestamp": "11:09:52"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 5, "pressure_psi": 86.5, "flow_rate_g
pm": 45.0, "timestamp": "11:09:53"}
^C
```

```
1776881389: Sending CONNACK to experiment-publisher (0, 0)
1776881389: Received PUBLISH from experiment-publisher (d0, q1, r0, m1, 'grandmarina/sensors/test/telemetry', ... (120 b
ytes))
1776881389: Sending PUBLISH to auto-0A3B4FCC-5E59-B55E-2D40-D241A4091EB0 (d0, q0, r0, m0, 'grandmarina/sensors/test/tele
metry', ... (120 bytes))
1776881389: Sending PUBACK to experiment-publisher (m1, rc0)
1776881390: Received PUBLISH from experiment-publisher (d0, q1, r0, m2, 'grandmarina/sensors/test/telemetry', ... (120 b
ytes))
1776881390: Sending PUBLISH to auto-0A3B4FCC-5E59-B55E-2D40-D241A4091EB0 (d0, q0, r0, m0, 'grandmarina/sensors/test/tele
metry', ... (120 bytes))
1776881390: Sending PUBACK to experiment-publisher (m2, rc0)
1776881391: Received PUBLISH from experiment-publisher (d0, q1, r0, m3, 'grandmarina/sensors/test/telemetry', ... (120 b
ytes))
1776881391: Sending PUBLISH to auto-0A3B4FCC-5E59-B55E-2D40-D241A4091EB0 (d0, q0, r0, m0, 'grandmarina/sensors/test/tele
metry', ... (120 bytes))
1776881391: Sending PUBACK to experiment-publisher (m3, rc0)
1776881392: Received PUBLISH from experiment-publisher (d0, q1, r0, m4, 'grandmarina/sensors/test/telemetry', ... (120 b
ytes))
1776881392: Sending PUBLISH to auto-0A3B4FCC-5E59-B55E-2D40-D241A4091EB0 (d0, q0, r0, m0, 'grandmarina/sensors/test/tele
metry', ... (120 bytes))
1776881392: Sending PUBACK to experiment-publisher (m4, rc0)
1776881393: Received PUBLISH from experiment-publisher (d0, q1, r0, m5, 'grandmarina/sensors/test/telemetry', ... (120 b
```

Images shown above are with TLS off, now we turn it on.

```
(base) C:\Users\nsain\Documents\hydroficient-project\Project-04>python experiment_runner.py --mode publish --tls on --co
unt 5

==================================================
  Publishing 5 messages
  TLS: ON
  Port: 8883
==================================================

Published message 1/5: pressure=82.5 PSI
Published message 2/5: pressure=83.5 PSI
Published message 3/5: pressure=84.5 PSI
Published message 4/5: pressure=85.5 PSI
Published message 5/5: pressure=86.5 PSI

Done publishing!
```

```
1776881570: Received PUBLISH from experiment-publisher (d0, q1, r0, m1, 'grandmarina/sensors/test/telemetry', ... (120 b
ytes))
1776881570: Sending PUBACK to experiment-publisher (m1, rc0)
1776881571: Received PUBLISH from experiment-publisher (d0, q1, r0, m2, 'grandmarina/sensors/test/telemetry', ... (120 b
ytes))
1776881571: Sending PUBACK to experiment-publisher (m2, rc0)
1776881572: Received PUBLISH from experiment-publisher (d0, q1, r0, m3, 'grandmarina/sensors/test/telemetry', ... (120 b
ytes))
1776881572: Sending PUBACK to experiment-publisher (m3, rc0)
1776881573: Received PUBLISH from experiment-publisher (d0, q1, r0, m4, 'grandmarina/sensors/test/telemetry', ... (120 b
ytes))
1776881573: Sending PUBACK to experiment-publisher (m4, rc0)
1776881574: Received PUBLISH from experiment-publisher (d0, q1, r0, m5, 'grandmarina/sensors/test/telemetry', ... (120 b
ytes))
1776881574: Sending PUBACK to experiment-publisher (m5, rc0)
1776881575: Received DISCONNECT from experiment-publisher
1776881575: Client experiment-publisher [::1:26496] disconnected.
```

```
(base) C:\Users\nsain\Documents\hydroficient-project\Project-04>mosquitto_sub -h localhost -p 8883 -t "grandmarina/#" -v

Error: protocol error
```

- This shows that the eavesdropper (me) has been denied due to TLS being on.

- ***Experiment 2/4*** *will involve fake certificates, the overall test is to see if an attacker's broker with a fake certificate is getting signed by our CA and giving the client access to the wrong broker.*
- **Scenario A:** First we check if the root certificate is valid, here I will run `python experiment_runner.py --mode connect --tls on` This shows a successful connection

```
(base) C:\Users\nsain\Documents\hydroficient-project\Project-04>python experiment_runner.py --mode connect --tls on

==================================================
  Connection Test
  TLS: ON
  CA Certificate: certs/ca.pem
==================================================

SUCCESS: Connected to broker!
```

- **Scenario B:** Generating a false CA Certificate, This test shows how TLS is preventing MitM attacks (This certificate was rejected by the client) in this case if our certificate was completely off, anyone would be able to connect with their certificate.

```
(base) C:\Users\nsain\Documents\hydroficient-project\Project-04>python experiment_runner.py --mode  test-wrong-ca
Testing connection with wrong CA...

==================================================
  Connection Test
  TLS: ON
  CA Certificate: certs/wrong-ca.pem
==================================================

FAILED: [SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed: self-signed certificate in certificate chain (_ssl.c
:1032)
```

- **Experiment 3/4:** This experiment involves checking the latency for the waterpipe sensors. I have to see how much latency TLS adds and is it crucial is the question. If a water leak was to happen I would need a fast response, This experiment will show that.

TLS OFF

```
(base) C:\Users\nsain\Documents\hydroficient-project\Project-04>python experiment_runner.py --mode latency --tls off --count 50

==================================================
 Latency Test
 TLS: OFF
 Messages: 50
==================================================

 Sent 10/50 messages...
 Sent 20/50 messages...
 Sent 30/50 messages...
 Sent 40/50 messages...
 Sent 50/50 messages...

==================================================
 Latency Results (TLS OFF)
==================================================
 Messages sent: 50
 Average latency: 50.95 ms
 Min latency: 0.32 ms
 Max latency: 100.83 ms
 Std deviation: 49.94 ms
==================================================
```

TLS ON

```
(base) C:\Users\nsain\Documents\hydroficient-project\Project-04>python experiment_runner.py --mode latency --tls on --count 50

==================================================
 Latency Test
 TLS: ON
 Messages: 50
==================================================

 Sent 10/50 messages...
 Sent 20/50 messages...
 Sent 30/50 messages...
 Sent 40/50 messages...
 Sent 50/50 messages...

==================================================
 Latency Results (TLS ON)
==================================================
 Messages sent: 50
 Average latency: 52.19 ms
 Min latency: 0.39 ms
 Max latency: 100.83 ms
 Std deviation: 49.34 ms
```

Without TLS: 50.95 ms average

With TLS: 52.19 ms average

Overhead(52.19 - 50.95) / 50.95 x 100% = 2.4%

- These values are negligible, so this won't impact the sensor as it only runs every 2-5 seconds.

- **Experiment 4/4:** This experiment will be a stress test, how many messages at a time can the sensor handle. This experiment will consist of TLS being off and on for a message rate of **10/sec, 25/sec, 50/sec, 100/sec**

10/sec

```
==================================================
  Stress Test Results (TLS OFF)
==================================================
  Target rate: 10 msg/sec
  Actual rate: 9.7 msg/sec
  Messages sent: 300
  Errors: 0
  Success rate: 100.0%
  Status: SUCCESS (achieved target rate)
==================================================
==================================================
  Stress Test Results (TLS ON)
==================================================
  Target rate: 10 msg/sec
  Actual rate: 9.7 msg/sec
  Messages sent: 300
  Errors: 0
  Success rate: 100.0%
  Status: SUCCESS (achieved target rate)
==================================================
```

25/sec

```
==================================================
  Stress Test Results (TLS OFF)
==================================================
  Target rate: 25 msg/sec
  Actual rate: 24.2 msg/sec
  Messages sent: 750
  Errors: 0
  Success rate: 100.0%
  Status: SUCCESS (achieved target rate)
==================================================
==================================================
  Stress Test Results (TLS ON)
==================================================
  Target rate: 25 msg/sec
  Actual rate: 24.2 msg/sec
  Messages sent: 750
  Errors: 0
  Success rate: 100.0%
  Status: SUCCESS (achieved target rate)
==================================================
```

50/sec

```
==================================================
  Stress Test Results (TLS OFF)
==================================================
  Target rate: 50 msg/sec
  Actual rate: 48.4 msg/sec
  Messages sent: 1500
  Errors: 0
  Success rate: 100.0%
  Status: SUCCESS (achieved target rate)
==================================================
  Stress Test Results (TLS ON)
==================================================
  Target rate: 50 msg/sec
  Actual rate: 48.4 msg/sec
  Messages sent: 1500
  Errors: 0
  Success rate: 100.0%
  Status: SUCCESS (achieved target rate)
==================================================
```

100/sec

```
==================================================
  Stress Test Results (TLS OFF)
==================================================
  Target rate: 100 msg/sec
  Actual rate: 96.8 msg/sec
  Messages sent: 3000
  Errors: 0
  Success rate: 100.0%
  Status: SUCCESS (achieved target rate)
==================================================
==================================================
  Stress Test Results (TLS ON)
==================================================
  Target rate: 100 msg/sec
  Actual rate: 96.8 msg/sec
  Messages sent: 3000
  Errors: 0
  Success rate: 100.0%
  Status: SUCCESS (achieved target rate)
==================================================
```

The success rate was 100% throughout the Stress Test experiment, this shows that in normal operations 10msg/sec runs smoothly with TLS ON, It also runs in emergency mode 50msg/sec smoothly, therefore the experiment was a success and nothing was degraded.